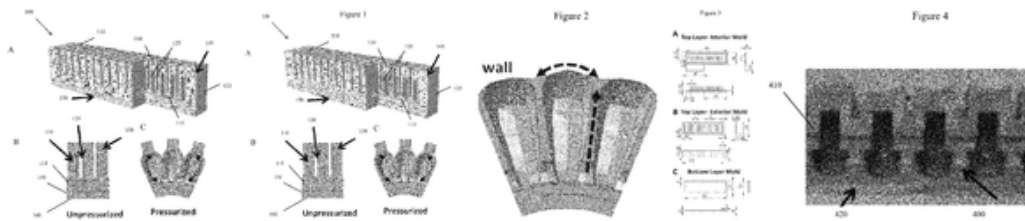


Low strain pneumatic networks for soft robots

Abstract

An actuator includes a plurality of chambers comprised of an extensible material, the chambers having interior side walls and exterior walls, wherein at least a portion of the interior side wall is separated from an interior side wall of an adjacent chamber; and a strain limiting base; and a channel that fluidically interconnects the plurality of chambers, wherein the interior walls are configured to be more compliant than the exterior walls.

Images (16)



Classifications

- **B25J15/0023** Gripper surfaces directly activated by a fluid

[View 7 more classifications](#)

Landscapes

Engineering & Computer Science

Mechanical Engineering

Robotics

US10639801B2

United States

Download PDF

Find Prior Art

Similar

Inventor: Bobak Mosadegh, Robert F. Shepherd, George M. Whitesides

Current Assignee : Harvard University

Worldwide applications

2014 ~~WE~~ US

Application US14/464,396 events 

2014-08-20 Application filed by Harvard University

2014-08-20 Priority to US14/464,396

2015-04-27 Assigned to PRESIDENT AND FELLOWS OF HARVARD COLLEGE ?

2015-09-24 Publication of US20150266186A1

2020-05-05 Application granted

2020-05-05 Publication of US10639801B2

Status	Active
--------	--------

2037-12-20 Adjusted expiration

Info: Patent citations (6), Non-patent citations (14) , Cited by (37), Legal events, Similar documents, Priority and Related Applications

Physics & Mathematics

Fluid Mechanics

General Engineering & Computer Science

Chemical & Material Sciences

Analytical Chemistry

Actuator

External links: [USPTO](#), [USPTO PatentCenter](#), [USPTO Assignment](#), [Espacenet](#), [Global Dossier](#), [Discuss](#)

Show less

Claims (32)

Hide Dependent ^

The invention claimed is:

1. An actuator, comprising:

a base comprising a strain limiting layer;

a plurality of unsupported chambers secured to the strain limiting layer, each chamber having interior side walls, exterior side walls, and a top wall, wherein the unsupported chambers do not share a common wall along a portion of the interior side walls;

a channel that fluidically interconnects the plurality of chambers; and

an inlet port for receiving a pressurizing fluid, wherein the interior side walls expand a greater amount relative to the exterior side walls and the top wall upon pressurization of the chambers to cause bending of the actuator around the strain limiting layer and cause said portion of the interior walls to touch each other.

2. The actuator of claim 1, wherein the exterior walls include exterior side walls having a thickness that is greater than a thickness of the interior side walls.

3. The actuator of claim 1, wherein the exterior walls include top walls having a thickness that is greater than a thickness of the interior side walls.

4. The actuator of claim 1, wherein the exterior walls are made from a materials having a greater stiffness than the interior side walls.

5. The actuator of claim 1, wherein the strain limiting layer comprises an inextensible, flexible layer.

6. The actuator of claim 5, wherein the strain limiting layer comprises an inextensible layer embedded in a layer of flexible polymer.

7. The actuator of claim 6, wherein the inextensible layer comprises a paper, a fabric or a polymer film.
8. The actuator of claim 7, wherein the paper or the fabric is provided as a matted sheet, a woven sheet or a fiber aligned along the length of the strain limiting layer.
9. The actuator of claim 1, wherein the strain limiting layer comprises an extensible, flexible material that is configured and arranged to be stiffer than the exterior walls of the chambers.
10. The actuator of claim 1, wherein the plurality of chambers are comprised of an elastomer.
11. The actuator of claim 1, wherein the actuator is powered by pressure provided by a gas or liquid source.
12. The actuator of claim 1, wherein the channel is positioned between the interior sidewalls of the plurality of chambers.
13. The actuator of claim 1, wherein the channel is located in the strain limiting layer and is positioned and arranged to fluidically connect the plurality of chambers.
14. The actuator of claim 1, wherein greater than 25% of the surface area of the interior side wall of the first chamber is separated from the interior side wall of the second chamber.
15. A method of actuation comprising:
 - providing the actuator according to claim 1; and
 - pressurizing the actuator to cause the actuator to actuate from a first resting position to a second actuated position.
16. The method of claim 15, wherein pressurizing causes the interior side walls to expand preferentially compared to the exterior walls.
17. The method of claim 15, wherein the actuation occurs in less than 1 second.
18. The method of claim 15, wherein the actuation occurs in less than 250 msec.
19. The method of claim 15, wherein the actuation occurs in less than 100 msec.
20. The method of claim 15, wherein the actuation is greater than 1 m/sec.
21. The method of claim 15, wherein the actuator undergoes a circumferential-expansion of less than 3% during the actuation.
22. The method of claim 15, wherein the actuator undergoes an energy loss of less than 10% between the actuated and unactuated state.
23. The method of claim 15, wherein the actuator is configured to be cycled more than 10000 times without failure.
24. The method of claim 15, wherein the actuator is configured to be cycled more than 200000 times without failure.
25. The method of claim 15, wherein the actuator is configured to cycle more than one million times without failure.
26. The method of claim 15, wherein the frequency of actuation is greater than 1 Hz.
27. The method of claim 15, wherein the frequency of actuation is greater than 2 Hz.

28. The method of claim 15, wherein the actuator bends along a circular trajectory below a threshold actuation rate and the tip of the actuator bends preferentially, causing the actuator to curl on itself above a threshold actuation rate.

29. A soft robot comprising:

a plurality of soft actuators according to claim 1; and

a connector having a plurality of fluidic inputs and a pressurizing inlet, wherein the pressurizing inlet is configured to receive fluid, and the inlet port of one of the plurality of soft actuators is reversibly coupled to one of the plurality of fluidic inputs.

30. The actuator of claim 1, wherein the inlet port is coupled to a solenoid valve that is configured to control whether the actuator is pressurized with compressed air or vented to atmosphere.

31. The actuator of claim 30, wherein the solenoid valve is coupled to the inlet port via tubing.

32. The actuator of claim 1, wherein the plurality of chambers are secured to the strain limiting layer along one face.

Description

CROSS-REFERENCE TO RELATED APPLICATION

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

INCORPORATION BY REFERENCE

TECHNICAL FIELD

BACKGROUND

SUMMARY

In one or more embodiments, the exterior walls include exterior side walls having a thickness that is greater than the thickness of the interior side walls.

In one or more embodiments the exterior walls include top walls having a thickness that is greater than a thickness of the interior side walls.

In any of the preceding embodiments, the base can include an inextensible, flexible layer.

In any of the preceding embodiments, the base can include an inextensible layer embedded in a layer of flexible polymer, and for example, the inextensible layer can be paper, fabric or polymer film, and for example, the paper or fabric is provided as a matted sheet, woven sheet or fibers aligned along the length of the base.

BRIEF DESCRIPTION OF THE DRAWINGS

DETAILED DESCRIPTION

The low strain, rapid actuation pneumatic network includes an extensible top layer and inextensible bottom layer. The soft robots include a plurality of unsupported chamber or “freestanding” chambers secured to a common strain limiting layer along one face. The unsupported chamber does not share a common wall along the at least a portion of the length of the side wall. The chambers can be “detached”, that is, the chamber does not share any portion of its side wall with its neighboring chambers, or “semi-detached”, that is, a chamber shares a portion of its side wall with its neighboring chambers. In other embodiments, the pneumatic network can be described as being a ‘double-contoured’ pneumatic network, because the contour of the upper surface and lower surface are different. The chambers are fluidically connected, for example, through a pneumatic conduit located in the base containing the strain limiting layer or in the side walls of the chambers. Soft robots possessing a plurality of detached, semi-detached or freestanding chambers secured to a common strain limiting layer along one face can be pressurized to provide large and versatile actuation to soft elastomers. The interior side walls are designed to be more compliant than the exterior walls. In one or more embodiments, the exterior walls are thicker than the interior side walls. In one or more embodiments, the exterior walls are made of a stiffer material than the interior side wall to obtain a more compliant interior side walls. Therefore, an increase in the internal pressure preferentially expands the inside walls and minimizes the strain that occurs on the other exterior walls. In addition, the close proximity of two neighboring chambers causes the expanding inside walls to push against each other, and results in a preferential elongation of the extensible layer, with only small changes (<1%) in the height of this layer. While described below for use with pneumatic activation, it is recognized that the device may be used with hydraulic activation. The response times may be longer for hydraulic activation.

The extensible layer is connected to a strain limiting base **150** including a stiffer, yet still pliable backing layer **160**. The stiffer backing layer may be a material having a higher elastic modulus than the chambers. The high modulus material can be used for the base where inflation is undesirable, while a low elastic modulus material can be used for the chambers where extensibility is needed. Alternatively, the stiffer backing layer may include an inextensible backing material, such as paper or fabric, embedded in the base. In such cases, the base and chambers can be made of the same or different materials.

The size and shape of the chambers can be varied. Variable that can be varied include chamber height, thickness of the interior side walls, thickness of the top and exterior side walls, and number of chambers in the pneumatic actuator. Generally, increasing the number of chambers increases the curvature of actuation at a given pressure (due to additive effect of actuation of each individual chamber). Increasing chamber height reduces curvature of actuation at a given pressure. The chamber features can be curved, rounded or rectilinear.

A variety of chamber design parameters can be modified in order to change the actuation response in the pneumatic network. Exemplary parameters that can be varied to control the performance of the freestanding actuator include the number, height, and wall thickness of the chambers. The effect of these parameters on the performance of the actuator was evaluated by measuring the radius of curvature at 4 PSI and the pressure required to actuate the low strain pneumatic network to its full range of motion. The results are shown in FIGS. 5A-5C. FIGS. 5A-5C show curvature of the actuator when pressurized with 4 PSI for pneumatic actuators prepared using freestanding chambers having different chamber heights, wall thickness and number of chambers. FIGS. 5D-5F show the pressure required for each of these designs to be fully actuated. For FIGS. 5A and 5D, an actuator with height of zero is a point of comparison with an actuator having embedded chambers as illustrated in FIG. 2. An actuator is considered fully actuated when both ends of the actuator were in contact. A pressure gauge was connected to the actuator using tubing to provide a reading of internal pressure.

To demonstrate the precision and speed of the low strain pneumatic networks according to one or more embodiments, we used four actuators to play “Mary Had a Little Lamb” on the keyboard, which is shown in FIG. 15A. In some embodiments, the four actuators having freestanding chambers can be affixed on the keyboard using Velcro and can be actuated with air using computer-regulated solenoid valves. In some embodiments, the actuators can be coupled together using a connector. The connector can include a plurality of fluidic inputs and a pressurizing inlet. The pressurizing inlet can be configured to receive fluid, and one of the actuators can be reversibly attached to one of the plurality of fluidic inputs. Each note was played by pressurizing one of the actuators for 75 ms with a head pressure of 15 psi, and then venting them for 150 ms. For notes played in succession (the same actuator pressurized twice), we used a 50 ms pressurization time so the actuator would release the key more quickly (alternatively, a longer venting time could have been used but would have resulted in a slower tempo).

Patent Citations (6)

Publication number	Priority date	Publication date	Assignee	Title
US4784042A *	1986-02-12	1988-11-15	Nathaniel A. Hardin	Method and system employing strings of opposed gaseous-fluid inflatable tension actuators in jointed arms, legs, beams and columns for controlling their movements
US20020157388A1	2000-03-28	2002-10-31	Takeshi Seto	Pump-integrated flexible actuator
US20030110938A1 *	2001-12-13	2003-06-19	Seiko Epson Corporation	Flexible actuator
US20120216672A1	2009-10-07	2012-08-30	Carlo Menon	Fluidic actuator and method of manufacture
WO2012148472A2	2010-11-19	2012-11-01	President And Fellows Of Harvard College	Soft robotic actuators
WO2013110086A1	2012-01-19	2013-07-25	President And Fellows Of Harvard College	Flexible robotic actuators
Family To Family Citations				

* Cited by examiner, † Cited by third party

Non-Patent Citations (14)

Title

Brown, E., et al., Universal robotic gripper based on the jamming of granular material. Proceedings of the National Academy of Sciences, 2010, 5 pages.

Chou, C.P. and B. Hannaford, Measurement and modeling of McKibben pneumatic artificial muscles. IEEE Transactions on Robotics and Automation, 1996. 12(1): p. 90-102.

Ilievski, F., et al., Soft robotics for chemists. Angew Chem Int Ed Engl, 2011. 50(8): p. 1890-5.

International Search Report and Written Opinion dated Apr. 16, 2015, in the International application No. PCT/US14/51818, filed Aug. 20, 2014, 13 pages.

Kim, M.S., et al., Manufacturing of inchworm robot using shape memory alloy (SMA) embedded composite structure. International Journal of Precision Engineering and Manufacturing, 2011. 12(3): p. 565-568.

Lee, H., C.G. Xia, and N.X. Fang, First jump of microgel; actuation speed enhancement by elastic instability. Soft Matter, 2010. 6(18): p. 4342-4345.

Martinez, R.V., et al., Elastomeric Origami: Programmable Paper-Elastomer Composites as Pneumatic Actuators. Advanced Functional Materials, 2012. 22(7): p. 1376-1384.

Martinez, R.V., et al., Robotic tentacles with three-dimensional mobility based on flexible elastomers. Adv Mater, 2013. 25(2): p. 205-12.

Morin, S.A., et al., Camouflage and display for soft machines. Science, vol. 337, No. 6096, Aug. 17, 2012, pp. 828-832.

Otake, M., et al., Motion design of a starfish-shaped gel robot made of electro-active polymer gel. Robotics and Autonomous Systems, 2002. 40(2-3): p. 185-191.

Shepherd, R.F., et al., Multigait soft robot. Proc Natl Acad Sci U S A, 2011. 108(51): p. 20400-3.

Shepherd, R.F., et al., Using Explosions to Power a Soft Robot. Angew Chem Int Ed Engl, vol. 52, pp. 2892-2896, 2013.

Shi, L.W., et al., A Novel Soft Biomimetic Microrobot with Two Motion Attitudes. Sensors, 2012. 12(12): p. 16732-16758.

Wakimoto, S., K. Suzumori, and K. Ogura, Miniature Pneumatic Curling Rubber Actuator Generating Bidirectional Motion with One Air-Supply Tube. Advanced Robotics, 2011. 25(9-10): p. 1311-1330.

* Cited by examiner, † Cited by third party

Cited By (37)

Publication number	Priority date	Publication date	Assignee	Title
US20240209877A1 *	2022-12-21	2024-06-27	Toyota Motor Engineering & Manufacturing North America, Inc.	Inflatable structure with reinforcement shaped according to a turing pattern
Family To Family Citations				

JP6721515B2 *	2014-04-11	2020-07-15	プレジデント アンド フェローズ オブ ハーバード カレッジ	Portable prosthetic hand with soft pneumatic fingers
US10006444B2 *	2014-04-11	2018-06-26	President And Fellows Of Harvard College	High throughput fabrication of soft machines
WO2016176340A2 *	2015-04-27	2016-11-03	Regents Of The University Of Minnesota	Soft robots, soft actuators, and methods for making the same
WO2017011438A1 *	2015-07-13	2017-01-19	President And Fellows Of Harvard College	Distributed pressurization and exhaust systems for soft robots
US9802314B2 *	2015-10-01	2017-10-31	Disney Enterprises, Inc.	Soft body robot for physical interaction with humans
US10350504B2	2016-09-13	2019-07-16	Universal City Studios Llc	Systems and methods for incorporating pneumatic robotic systems into amusement park attractions
CN107914269B *	2016-10-09	2020-01-03	中国科学技术大学	Software robot based on honeycomb pneumatic network
US11918529B2 *	2017-01-23	2024-03-05	National University Of Singapore	Fluid-driven actuator and its applications
DE102018103178A1 *	2018-02-13	2019-08-14	Baden-Württemberg Stiftung Ggmbh	Component with integrated actuator
US11103991B2	2018-03-21	2021-08-31	Arizona Board Of Regents On Behalf Of Arizona State University	Assisted lifting devices
CN108297955B *	2018-03-29	2023-07-04	燕山大学	A bionic wheel-footed peristaltic soft robot
CN108556234B *	2018-05-11	2019-06-04	清华大学	A mold and method for manufacturing a soft robot
KR102133497B1 *	2018-06-05	2020-07-13	울산과학기술원	Soft gripper unit and soft robot apparatus having the same
US10456929B1 *	2018-08-13	2019-10-29	Schlumberger Technology Corporation	Negative pressure actuated soft bending actuator
CN108927800B *	2018-09-06	2023-09-29	江南大学	Roll blowing type quick telescopic robot and motion control mode thereof
KR101950654B1 *	2018-12-21	2019-02-20	서울대학교산학협력단	Soft block unit comprising expanding block and bending block
CN109648545A *	2019-01-31	2019-04-19	江苏大学	A kind of soft robot based on hydrogel driver
CN109968386A *	2019-05-17	2019-07-05	江南大学	A flexible gripper with surface microstructure

US10973528B2	2019-05-29	2021-04-13	The Government of the United States of America, as represented by the Secretary of Homeland Security	Modular system and apparatus for treating wounds
CN110104083B *	2019-06-05	2021-04-13	苏州柔性智能科技有限公司	Software of many motion modes robot of crawling
CN110394794A *	2019-06-27	2019-11-01	西安工业大学	A kind of software driving device
CN110478192A *	2019-08-30	2019-11-22	哈工大机器人湖州国际创新研究院	Soft auxiliary glove and method of making the same
CN110539292B *	2019-10-09	2024-05-28	北京化工大学	Semi-soft robot module with rigidity retention and semi-soft robot
US11530621B2	2019-10-16	2022-12-20	General Electric Company	Systems and method for use in servicing a machine
KR102252689B1 *	2019-10-29	2021-05-18	중앙대학교 산학협력단	Light Weight and High Bending Angle Layer Based Bending Actuator
CN111975808A *	2020-07-08	2020-11-24	河海大学常州校区	Air control soft bionic mechanical finger
CN114407004A *	2022-02-18	2022-04-29	广州大学	Pneumatic flexible actuator
CN114619457B	2022-03-23	2023-08-01	东南大学	A dual-airway bionic snail soft robot
CN114734433A *	2022-04-07	2022-07-12	深圳大学	Multi-joint soft manipulator and manufacturing method thereof
CN116872657A *	2022-08-23	2023-10-13	北京大学	A soft amphibious robot
KR102817848B1 *	2023-01-13	2025-06-09	서울대학교산학협력단	Snap-through joint and soft robot including same
CN116652991A *	2023-06-05	2023-08-29	天津理工大学	A tendril-like variable-curvature soft actuator
CN116788380B *	2023-06-28	2025-12-02	南开大学	A crocodile-like soft-crawling robot, its control method and system
CN117601122B *	2023-11-23	2024-04-30	哈尔滨工业大学	A pneumatic artificial muscle position tracking control system and method
CN120080311B *	2025-04-07	2025-11-07	中国矿业大学(北京)	Pneumatic soft driver and processing die
CN120134344A *	2025-05-15	2025-06-13	湖南大学	A logarithmic spiral bidirectional bending pneumatic soft actuator and soft robot

* Cited by examiner, † Cited by third party, ‡ Family to family citation

Similar Documents



Publication	Publication Date	Title
US10639801B2	2020-05-05	Low strain pneumatic networks for soft robots
US10385886B2	2019-08-20	Soft actuators and soft actuating devices
Mosadegh et al.	2014	Pneumatic networks for soft robotics that actuate rapidly
KR102084280B1	2020-03-03	Systems and methods for providing flexible robotic actuators
Lalegani Dezaki et al.	2023	Soft pneumatic actuators with controllable stiffness by bio-inspired lattice chambers and fused deposition modeling 3D printing
Hammond et al.	2012	Soft tactile sensor arrays for micromanipulation
US9133864B2	2015-09-15	Fluidic actuator and method of manufacture
CN115000287A	2022-09-02	Large scale integration of haptic devices
US20100258362A1	2010-10-14	Actuator powered deformable soft-bodied autonomous platforms
EP3600793A2	2020-02-05	Servo-pneumatic control systems for soft robotic actuators
CN107263549A	2017-10-20	Robot and external force detection device
US10882195B2	2021-01-05	Method for making a soft actuator device
KR20220088865A	2022-06-28	Tactile presentation device, fluid control device, fluid control method and program
EP2908986A2	2015-08-26	Apparatus, systems, and methods for actuating pressurizable chambers
Vo et al.	2025	Soft, fiber-reinforced bellow actuators
Bui et al.	2023	Endurance tests for a fabric_reinforced inflatable soft actuator
Shorthose et al.	2021	Design of a multimaterial 3D-printed soft actuator with bi-directional variable stiffness
Park et al.	2018	Development of a hybrid gripper with soft material and rigid structures
US11132061B1	2021-09-28	Fluidic haptic device operable by user movement
Chen et al.	2024	Design and characterization of a pneumatic tunable-stiffness bellows actuator

Deimel	2017	Soft robotic hands for compliant grasping
Ko et al.	2022	Characterization of bending balloon actuators
Colucci	2025	Pneumatic Soft Robotic System for Actuation and Sensing
Yao et al.	2025	JAMMit! Monolithic 3D-Printing of a Bead Jamming Soft Pneumatic Arm
Wang et al.	2018	A dynamical model with order reduction for a fluidic bending-type soft actuator

Priority And Related Applications

Priority Applications (1)

Application	Priority date	Filing date	Title
US14/464,396	2013-08-20	2014-08-20	Low strain pneumatic networks for soft robots

Applications Claiming Priority (2)

Application	Filing date	Title
US201361867845P	2013-08-20	
US14/464,396	2014-08-20	Low strain pneumatic networks for soft robots

Legal Events

Date	Code	Title	Description
2015-04-27	AS	Assignment	Owner name: PRESIDENT AND FELLOWS OF HARVARD COLLEGE, MASSACHUSETTS Free format text: ASSIGNMENT OF ASSIGNORS INTEREST;ASSIGNORS:MOSADEGH, BOBAK;SHEPHERD, ROBERT F.;WHITESIDES, GEORGE M.;SIGNING DATES FROM 20140904 TO 20150427;REEL/FRAME:035503/0199 Owner name: PRESIDENT AND FELLOWS OF HARVARD COLLEGE, MASSACHU

			Free format text: ASSIGNMENT OF ASSIGNORS INTEREST;ASSIGNORS:MOSADEGH, BOBAK;SHEPHERD, ROBERT F.;WHITESIDES, GEORGE M.;SIGNING DATES FROM 20140904 TO 20150427;REEL/FRAME:035503/0199
2019-02-19	STPP	Information on status: patent application and granting procedure in general	Free format text: FINAL REJECTION MAILED
2019-06-20	STPP	Information on status: patent application and granting procedure in general	Free format text: DOCKETED NEW CASE - READY FOR EXAMINATION
2019-07-05	STPP	Information on status: patent application and granting procedure in general	Free format text: NON FINAL ACTION MAILED
2019-10-13	STPP	Information on status: patent application and granting procedure in general	Free format text: RESPONSE TO NON-FINAL OFFICE ACTION ENTERED AND FORWARDED TO EXAMINER
2019-12-16	STPP	Information on status: patent application and granting procedure in general	Free format text: NOTICE OF ALLOWANCE MAILED – APPLICATION RECEIVED IN OFFICE OF PUBLICATIONS
2020-04-15	STCF	Information on status: patent grant	Free format text: PATENTED CASE
2023-11-06	MAFP	Maintenance fee payment	Free format text: PAYMENT OF MAINTENANCE FEE, 4TH YR, SMALL ENTITY (ORIGINAL EVENT CODE: M2551); ENTITY STATUS OF PATENT OWNER: SMALL ENTITY Year of fee payment: 4

Data provided by IFI CLAIMS Patent Services